

Dual ordinal spectral ranking of journals and institutions

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Abstract

We develop a dual ordinal spectral ranking of journal and institution units that determines influence-per-work ranking scores via block citation matrices. There are four bibliometrically distinct structures connecting journals (J) and institutions (I): J to J (similar to the Clarivate Eigenvector score), I to I (the institutional counterpart), J+I to J+I (full 2-unit citation matrix), and JI to IJ (recirculating). Applied to the economics and commerce journal literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically. Journal rankings are stabilised by the inclusion of institutions, and both unit's rankings remain remarkably similar (Spearman score ρ 0.99) over the period 2000-2024.

Keywords: spectral ranking, dual journal-institution prestige, Katz-Hubbell resolvent

1 Introduction

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval and exploration. Institutions fund individual researchers, implement training, oversee compliance, establish research organisational units and maintain physical and information infrastructure. Although the status hierarchy that links journals and institutions is a pervasive aspect of the research system (Merton, 1973; McNamee and Willis, 1994; Martin, 1998), bibliometric studies that explore the joint hierarchy of journals and institutions are infrequent. This paper proposes a joint ordinal spectral ranking of journals and institutions derived from bibliometric data. The method helps to reveal how status is formed and maintained. We intend to apply the approach to study the relations between the status of journals and institutions, and researcher esteem in later work.

The introduction of the science citation index (Garfield, 1964) opened a new way to investigate connections between journals and institutions. Citation-based journal ranking in particular became a useful means to manage collections and to select outlets for publications. On the other hand, the misapplication of the journal impact factor (JIF) as proxy for the importance of research works published in that journal fostered a still-unresolved tension in bibliometric evaluation (Garfield, 1979; MacRoberts and MacRoberts, 2018). Perhaps due to the complexity of their disciplines and missions, an institution impact factor (IIF) has not emerged although the Scimago Institution and. Rather, institutional rankings derived from bibliometric indicators of works, authors or journals are often composed with other indicators to form an institutional score. This is the approach in global rankings such as QS, Times Higher Education (THE), Shanghai (ARWU), Leiden Ranking (CWTS) and SCImago Institutions Rankings, as well as many national performance-based research funding schemes (Zacharewicz et al, 2019).

The prospects for using a citation index to explore research importance were considered in an early work by Narin (1976), Pinski and Narin (1976), and Narin et al (1976) who proposed a recursive method to construct influence weights of journals within a disciplinary field. To obtain influence weights for institutions, Narin summed the institutional publications in a field weighted by the respective journal influence score. Over a sample of several dozen universities, the total institution influence scores were well-correlated with survey-based rankings, but also with the number of publications, suggesting that survey respondents combined influence and size. Narin's work built on earlier studies of social networks and developed into the eigenvector approach now widely used for journal ranking (Bergstrom et al, 2008; Vigna, 2016; Franceschet, 2010). Eigenvector methods have also been applied to author-level influence weightings in the field of social science, and extended by Narin's approach to ranking institutions and countries (West et al, 2013). An alternative approach iterates between the article-article network and the corresponding article-author network (Ujum

et al, 2015; Pal and Ruj, 2015). The formulation by Cao et al (2023) adopts a similar network, with reference-based article-article links, and the related article-institution links, ranked by a novel and effective iteration over random walks. This is one of the few spectral rankings of institutions. The authors note that a limiting case of this approach corresponds to the eigenvector (PageRank) method: Geller (1978) noted that random walks over Markov chains are closely related to the spectral ranking introduced by Narin.

We foreground for readers some analytical devices adopted in this paper. Firstly, this paper views references and citations through the lens of Social Systems Citation Theory (Tahamtan and Bornmann, 2022, SSCT). It concentrates on the structural features of the network of references and citations, setting aside authorial, editorial and managerial intent. This approach contrasts with actor-network theory (Latour, 2007, ANT), which foregrounds the intentful roles of heterogeneous agents including authors in assembling citation networks. ANT will be important in our follow-up studies that relate this paper to the referencing activities of individual researchers. Second, we say that work w_1 *references* a work w_2 in the reference list of work w_1 , and that w_2 is *cited by* w_1 (Wouters, 1999). We say that a work w_1 pays or gives *attention* to its references, and that the references *influence* the work. We explain below how attention and influence at the work level are projected to journals and institutions.

The paper proceeds as follows. In the next section, we construct a framework for computing the transfer of attention or influence among journal and institution units coupled by the reference lists of the works they publish. Numerical methods for computing these ordinal spectral rankings are presented and their convergence discussed. We then construct a corpus of works in the fields of Economics and Business, forming subsets of the corpus to illustrate the interconnections of units, fields, and time. We present spectral rankings over these subsets, alongside sensitivity and bootstrap studies of some key parameters. The paper discusses these results including a critique of our methods, and concludes with some directions for future work.

2 Journal–institution citation networks

We begin by selecting N_s sources $\mathfrak{S}$ in a set of fields, disciplines or categories $\mathfrak{F}$. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $t_c = [t_0^c \dots t_1^c]$ or target years $t_t = [t_0^t \dots t_1^t]$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{r(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of institutions $\mathfrak{U}'$ and authors $\mathfrak{A}'$. In practice many institutions (and sources) publishing only a few works per year can be dropped from the corpus. Let $\bar{w}_u$ be the mean annual census-window work count for institution u and $\bar{w}_s$ the corresponding work count for source s . We determine lower limits τ_U and τ_S by examining the respective frequency distributions and define $\mathfrak{U} \subseteq \mathfrak{U}'$ as the institutions with $\bar{w}_u \geq \tau_U$, and $\mathfrak{S} \subseteq \mathfrak{S}'$ as the sources with $\bar{w}_s \geq \tau_S$, with corresponding author set $\mathfrak{A} \subseteq \mathfrak{A}'$. Works that lose all their authorships under the institution filter, or whose source is dropped by the source filter, are removed from the corpus. The choice to apply $\tau_{S,I}$ once across the corpus or else within each time window is examined below.

Let $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$ be the reference count of work w_i and $\bar{R} = |\mathfrak{R}|/N_w$ the mean reference count over $\mathfrak{W}$. Since reference lists have different lengths, we consider two cases for weighting references,

$$\rho_i = \begin{cases} 1 & \text{(full reference counting)} \\ \bar{R}/R_i & \text{(fractional reference counting)}. \end{cases} \quad (1)$$

Under full reference counting a work with twice as many references has twice the prominence; under fractional reference counting every work's references count equally (Zitt and Small, 2008).

A work has a retained source and one or more institutions. For journal spectral ranking, it is straightforward to link the work to its journal. In dual spectral ranking, however, we should consider (i) the fractionation between the journal and the institutions, and (ii) the fractionation across institutions. The parameter χ fractionates the source (χ) and institutions ($1 - \chi$). While at first sight the choice $\chi = 0.5$ appears sound, we see below that $\chi = N_S/(N_S + N_I)$ arises naturally once all works have been accumulated.

There are two models fractionating a work’s institutions (i.e. the *institution weight*)

$$\omega_{iu} = \begin{cases} \sum_{\ell: u \in \Pi_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, & \text{(author-fractional)} \\ \frac{1}{n_i}, & \text{(institution-fractional)}. \end{cases} \quad (2)$$

Here, $n_i = |\bigcup_{\ell} \Pi_{i\ell}|$ is the number of institutions of work w_i , ℓ indexes the authors of work w_i , a_i is their count, and $u_{i\ell}$ the number of institutions of author ℓ . Note that $\sum_u \omega_{iu} = 1$ under author-fractional counting (Waltman and van Eck, 2015). Institution pruning due to annual output falling below τ_u precedes fractional counting, a trade-off between boosting a work’s remaining institutions and diminishing the work’s total contribution.

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{U}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \dots, N_s$ and institutions indices $N_s + 1, \dots, N$. Each reference $r(i, j) \in \mathfrak{R}$ connects units associated with w_i to units associated with w_j , across four unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (2). The vectors $\mathbf{b}$ yield the projection across unit pairs, accumulated as four block matrices over $\mathfrak{R}$ via rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \rho_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}. \quad (3)$$

Here, $\mathbf{b}_i^X$ distributes the citing side across units of type X , and $\mathbf{b}_j^Y$ distributes the cited side across units of type Y . Each entry $[\mathbf{C}_{XY}]_{pq}$ is therefore proportional to the sum over $\mathfrak{R}$ of ρ_i -weighted fractional references from unit p of type X to unit q of type Y .

The citation matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \rho) = \begin{bmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{bmatrix}. \quad (4)$$

The binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$ is bibliometrically meaningful for source-only (1000), institution-only (0001), bipartite (0110), and full-joint (1111) combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. The χ scaling preserves the total weighted references across all block combinations, since $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$.

Note that the corpus excludes (a) census work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

Ordinal spectral ranking. Let $r_p = \sum_q C_{pq}$ be the sum of fractionated references given by unit p and $\mathbf{D}_r = \text{diag}(r_p)$ the corresponding diagonal matrix. Row-normalising $\mathbf{C}$ to obtain the row semi-stochastic matrix $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$ ensures that, when $\rho_i = 1$, reference intensity variations are removed. They are reintroduced later via the influence-per-work scaling (7). If $r_p = 0$ (a dangling unit with no outgoing references) the corresponding row of $\mathbf{H}$ is defined to be zero.

The three unipartite masks—source-only (1000), institution-only (0001), and full-joint (1111)—each yield a square $\mathbf{H}$ with the spectral ranking $\boldsymbol{\pi}^*$ (see Vigna, 2016, for an overview)

$$(\mathbf{I} - \mathbf{H}^\top) \boldsymbol{\pi}^* = \mathbf{0}, \quad \mathbf{1}^\top \boldsymbol{\pi}^* = 1, \quad \boldsymbol{\pi}^* \geq 0. \quad (5)$$

If $\mathbf{H}$ is primitive (irreducible and aperiodic) the ranking is unique due to the Perron–Frobenius theorem. Our numerical implementation tests primitivity for every $\mathbf{H}$, pre-processed as explained below, and has not encountered an imprimitive case. Geller (1978) showed that $\boldsymbol{\pi}^*$ is the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference count proportion $r_p / \sum_q r_q$.

Katz (1953) proposed attenuating walks (i.e. chains of citation matrix weights $c_{ab} \rightarrow c_{bc} \rightarrow c_{cd}$) of length k by a factor α^k , $\alpha \in (0, 1)$, so that distant links count for less than proximate ones. Summing walks of all lengths and endowing every unit with a uniform initial score of 1 yields the Katz resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = \mathbf{1}.$$

Since $\mathbf{H}$ is row-stochastic, $(\mathbf{I} - \alpha \mathbf{H}^\top)$ is invertible for any $\alpha \in (0, 1)$.

Hubbell (1965) generalised this by replacing the uniform endowment $\mathbf{1}$ with an informative prior $\boldsymbol{\mu}$ over units, representing exogenous information about standing independent of citation structure,

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = \boldsymbol{\mu}.$$

We explore two priors, (i) uniform over units $\mu_p = 1/N_{\mathfrak{P}}$ for $p \in \mathfrak{P}$ or (ii) uniform $1/(N_S + N_I)$. Note that the intermediate case $(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = \mathbf{0}$ with $\alpha \neq 1$ admits only the trivial solution $\boldsymbol{\pi} = \mathbf{0}$ and is not useful as a ranking.

Combining these yields the general (Katz-Hubbell) resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top) \boldsymbol{\pi} = (1 - \alpha) \boldsymbol{\mu}, \quad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \quad (6)$$

which recovers $\boldsymbol{\pi}^*$ in the limit $\alpha \rightarrow 1$ when $\mathbf{H}$ is primitive. While α and $\boldsymbol{\mu}$ are introduced here for their substantive interpretations—attenuation along walks and exogenous prior respectively—they have the additional effect of ensuring a unique solution to (6) even when $\mathbf{H}$ is not primitive: $\alpha < 1$ makes $(\mathbf{I} - \alpha \mathbf{H}^\top)$ invertible, and a nonzero $\boldsymbol{\mu}$ lifts the equation off the homogeneous system. If $\mathbf{C}$ were a binary matrix, this would be the PageRank spectral ranking (Brin and Page, 1998)

The spectral ranking $\boldsymbol{\pi}$ reflects the steady-state probability vector of the Markov matrix $\mathbf{H}$; we can call it the probability distribution of attention or influence. The *influence per work* is computed as (Palacios-Huerta and Volij, 2004),

$$\mathbf{v} = A \mathbf{A}^{-1} \boldsymbol{\pi}, \quad \mathbf{A} = \text{diag}(a_p), \quad A = \mathbf{a}^\top \mathbf{1} = \sum_p a_p. \quad (7)$$

Here, the unit work count $a_p = |\{i : w_i \in p\}|$ where p is a specific source or institution, and $\boldsymbol{\pi}$ denotes the solution to (6). The scalar A normalises $\mathbf{v}$ so that its a_p -weighted mean equals one: $\mathbf{a}^\top \mathbf{v} / A = \mathbf{1}^\top \boldsymbol{\pi} = 1$. A unit with $v_p = 1$ therefore has the corpus-average influence per work and $v_p = 2$ indicates twice the average influence per work. Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004).

Bipartite ranking. With $\mathbf{m} = (0110)$, equation (4) yields block off-diagonal matrices $\mathbf{C}$ and $\mathbf{H}$, forming a directed bipartite network between sources (S) and institutions (I). This mask is used as the baseline model in Section 4. Partitioning rows and columns into the S and I blocks, the semi-stochastic matrix takes the form

$$\mathbf{H} = \begin{pmatrix} \mathbf{0} & \mathbf{H}_{SI} \\ \mathbf{H}_{IS} & \mathbf{0} \end{pmatrix}, \quad (8)$$

where $\mathbf{H}_{SI}$ connects sources to institutions and $\mathbf{H}_{IS}$ does the reverse. Because every path alternates $S \rightarrow I \rightarrow S \rightarrow \dots$, this structure makes $\mathbf{H}$ periodic with period 2.

Squaring $\mathbf{H}$ collapses two-step paths to separate unit types, giving one-unit relations

$$\mathbf{H}^2 = \begin{pmatrix} \mathbf{M}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_I \end{pmatrix}, \quad \mathbf{M}_S = \mathbf{H}_{SI} \mathbf{H}_{IS}, \quad \mathbf{M}_I = \mathbf{H}_{IS} \mathbf{H}_{SI}.$$

Both $\mathbf{M}_S$ and $\mathbf{M}_I$ are row semi-stochastic products of row semi-stochastic matrices.

To make the bipartite mode comparable with the unipartite SS and II modes at the same α , single steps SI or IS should be damped by $\sqrt{\alpha}$, so the round-trip incurs attenuation $\sqrt{\alpha} \cdot \sqrt{\alpha} = \alpha$. Substituting $\sqrt{\alpha}$ for α in equation (5), partitioning $\boldsymbol{\pi} = (\boldsymbol{\pi}_S, \boldsymbol{\pi}_I)$, and eliminating $\boldsymbol{\pi}_I$ yields the bipartite resolvent for sources,

$$(\mathbf{I} - \alpha \mathbf{M}_S^\top) \boldsymbol{\pi}_S = (1 - \sqrt{\alpha}) (\boldsymbol{\mu}_S + \sqrt{\alpha} \mathbf{H}_{IS}^\top \boldsymbol{\mu}_I), \quad (9)$$

Note that when $\alpha < 1, \mu > 0$, the right-hand side links institutions to sources via $\mathbf{H}_{IS}^\top$. Institution ranking can be computed from source ranking by the matrix–vector product,

$$\boldsymbol{\pi}_I = \sqrt{\alpha} \mathbf{H}_{SI}^\top \boldsymbol{\pi}_S + (1 - \sqrt{\alpha}) \boldsymbol{\mu}_I. \quad (10)$$

Computation. Defining the modified matrix

$$\tilde{\mathbf{H}} = \alpha \mathbf{H} + (1 - \alpha) \mu \mathbf{1}^\top, \quad (11)$$

the resolvent $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}$ is equivalent to equation (6). This is the general form of our numerical problem, although most of the results reported below use $\alpha = 1$ and $\mu = 0$ so $\tilde{\mathbf{H}} = \mathbf{H}$.

As explained below, we construct many realisations of $\tilde{\mathbf{H}}$ while testing different models for unit pairings, subfields and other parameters, and to run our bootstrap investigations. All of these realisations entail a unit-unit citation network with one giant strongly connected component (SCC) and possibly some very small, usually singleton, components that are sources or institutions disjoint with the particular network under investigation. In these cases we retain only the SCC. For some units in the SCC, there may be no outgoing references, and for others, no incoming references. We exclude these units iteratively, ensuring that the residual $\tilde{\mathbf{H}}$ is stochastic and highly likely to be row-stochastic and primitive, with a single dominate eigenvector.

$\tilde{\mathbf{H}}$ is a sparse matrix constructed from the respective filtered and weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$ and stored in compressed sparse row (CSR) format. We have explored two approaches to applying the resolvent. The Python scipy function `scipy.sparse.linalg.eigs` efficiently provides the first few eigenpairs, allowing us to review the primitivity condition. The power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top \boldsymbol{\pi}^{(k)}, \quad (12)$$

converges to $\boldsymbol{\pi}$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$. In either approach, the per-work scaling equation (7) is applied after $\boldsymbol{\pi}$ has been computed.

3 Corpus selection

OpenAlex (Priem et al, 2022) provides our bibliometric data. Corpus selection starts with a list of eligible sources having three features: evidence from one of our journal registries that the source publishes economics or business research, an OpenAlex source identifier for corpus extraction, and a sufficient number of published works per year.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is confined to economics and business so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those having all FoR codes falling within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics). This applies to two-digit ‘division’ codes and all four-digit ‘group’ codes (3501–3509, 3801–3803, 3899). *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in Economics, Management, Business, Transportation or Business, Finance.

Every journal in JQL, SJL, and JCR has at least one ISSN, and may have several. These registry ISSNs are joined independently to the February 2026 OpenAlex sources snapshot using its list of ISSNs as the matching key and selecting only the ‘journal’ source-type. Journal titles are examined to identify around 30 additional matches where the OpenAlex source does not have an ISSN. The list of OpenAlex sources that are matched is designated OAS*.

For OAS* sources we extract work and authorship data to reveal some units that can be excluded because they publish very few works. To find a suitable threshold for excluding these units, we examine source and institution retention curves (Fig. 1), selecting thresholds of $\tau_U = 20$ works/institution/year and $\tau_S = 20$ works/source/year. We designate the residual set of sources OAS.

Using units that satisfy these thresholds, we extract works, authorships and references with publication years from 2000 to 2024, retaining article and review types and excluding works that are paratext or retracted, or without an authorship entity.

The sources cover the economics and business subfields within $\mathfrak{F}$, and it is of interest to compare spectral rankings for both. The three registries include one or more subfield indicators for each source, and OpenAlex sources have topic scores that indicate the relative intensity of business and economics subfields. Having

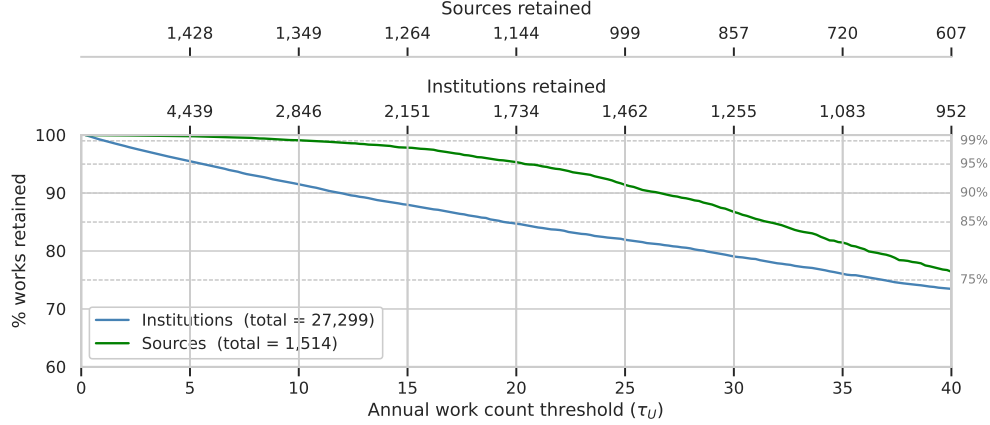


Figure 1: Institution and source retention curves

categorised subfields as ‘E’, ‘B’ or ‘None’ (e.g. Psychology or Education), we classify sources as ‘E’ when the summed number of indicators exceeds one ‘E’ mention with no ‘B’ mentions, ‘B’ with the complementary definition, ‘A’ when both subfields have more than one mention, and ‘X’ for sources don’t mention E or B. While sources are readily related to subfields, institutions generally publish substantial numbers of works in both subfields. Therefore, rather than use categorical classification, we classify institutions based on the ratio of their works in sources across the EBAX categories. The characteristics of the full corpus are exhibited in Table 1.

Table 1: Registry sources and their OpenAlex matches, classified by field. JQL = Harzing Journal Quality List; MJL = Web of Science Master Journal List; SJL = ERA Subject Journal List (FoR codes 35 and 38). A source appearing in multiple registries is counted in each. E/B/A/X are assigned by the combined four-signal scoring. OAS* is the union across all three registries matched to OpenAlex. OAS is OAS* restricted to sources with mean annual works $\geq \tau_S = 20$ in the census window 2020–2024.

Register	Sources	OA Match	Match%	E	B	A	X
JQL	842	835	99.2%	56	92	429	258
MJL	1,414	1,398	98.9%	163	165	782	288
SJL	1,603	1,379	86.0%	85	19	922	352
OAS*	2,294	—	—	251	196	1,006	841
OAS	1,685	—	—	158	157	786	584

4 Results

The results are organised into four themes: (i) unit-related, (ii) field-related, (iii) testing parameter effects, and (iv) bootstrap uncertainty estimates. Themes (i), (ii) and (iv) use the *baseline*, [2020–24, 2020–24], $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$, $\mu = 0$ — the eigenvector equation (5) on the bipartite network.

Figure 2 compares spectral ranks across network masks. The x -axis is the cardinal rank of sources (upper) and institutions (lower), with the solid black curve tracing the corresponding ordinal rank. The units near $v = 1$ lie at the weighted average influence-per-work, around rank position 25% for sources and 20% for institutions. Influence rises to > 10 for the top few units at the extreme left, with a low-influence tail falling below 0.1 for 10% of sources and $< 5\%$ for institutions. For sources, modes 1000 (red $\times$) and 1111 at χ^* (green $\times$) are overlaid on the bipartite baseline; open circles and squares on the 1000 series mark Economics and Business sources respectively. For institutions, mode 0001 and 1111 replace the source alternatives.

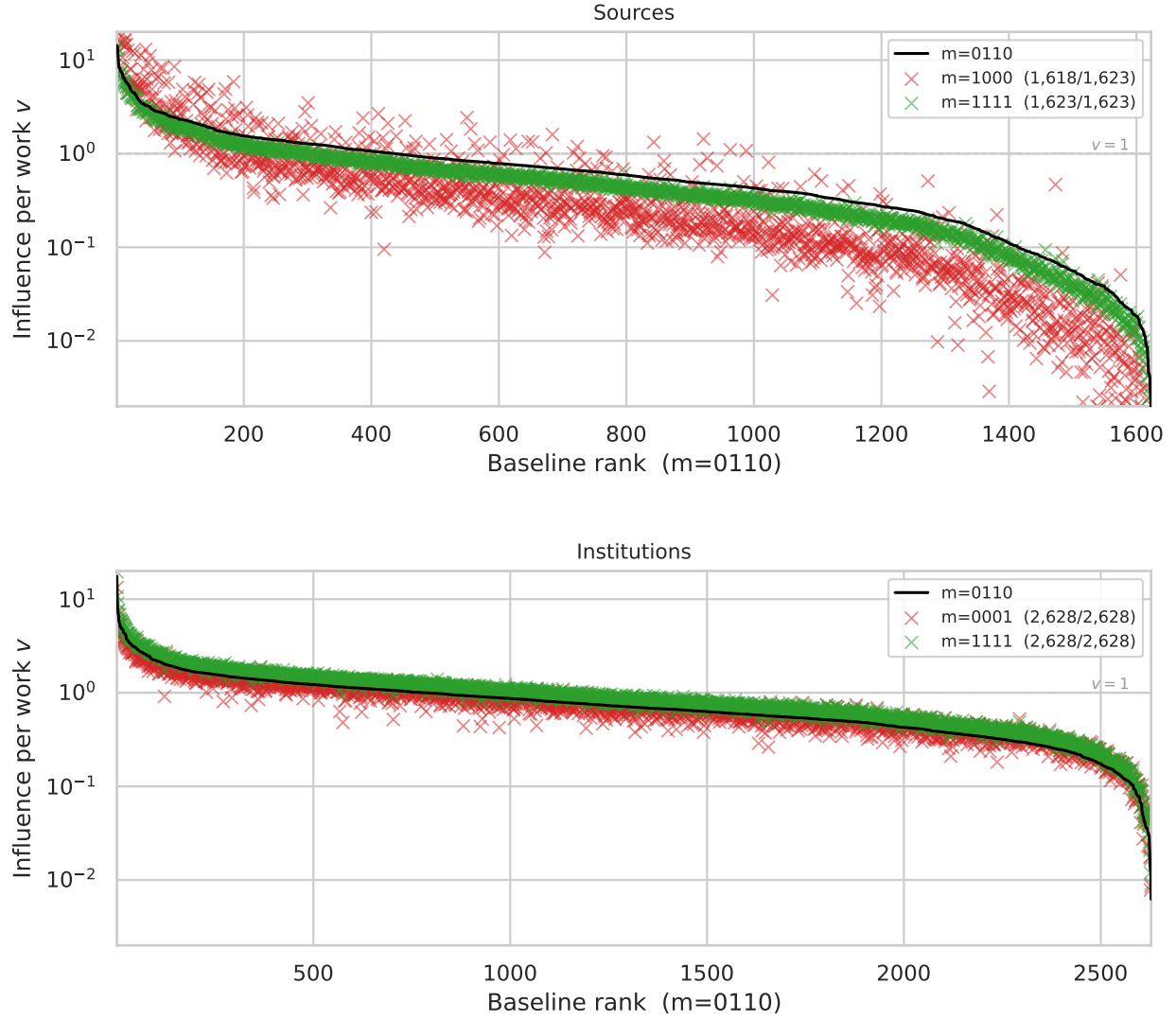


Figure 2: Prestige per work v against bipartite baseline rank. Top: sources. Bottom: institutions. Black line: bipartite baseline $\mathbf{m} = 0110$. Red $\times$: $\mathbf{m} = 1000$ (top) / $\mathbf{m} = 0001$ (bottom). Green $\times$: $\mathbf{m} = 1111$, $\chi^* = [FILL]$. Open circles ($\circ$) and squares ($\square$) overlaid on the 1000 series (top) and 0001 series (bottom) mark Economics and Business units respectively, classified by field-network membership. Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\alpha = 1$.

The most striking feature of Figure 2 is the contrast between the source and institution panels. For institutions, modes 0001 and 1111 track the bipartite baseline closely: prestige per work is largely insensitive to whether the institution-to-institution path is direct ($\mathbf{m} = 0001$) or mediated through sources ($\mathbf{m} = 0110$). For sources the picture is different: mode 1000 produces pronounced scatter both above and below the bipartite baseline, with a systematic downward shift for the majority and a distinct subset elevated well above it. The spectral gap $g = 1 - \lambda_2$ measures how quickly the random walk mixes across the network and thereby how strongly positional ranking depends on local neighbourhood rather than global structure. For the source-only mode $g^{SS} = 0.038$ —the walk on $\mathbf{H}_{SS}$ mixes very slowly, so a source’s rank reflects its position within its local citation cluster rather than its standing in the full network. Routing prestige through the SI – IS blocks dissolves this locality: the bipartite mode achieves $g^{0110} = 0.660$, nearly seventeen times larger, because institutions draw citations from and distribute prestige to sources across the citation network. The institution-only mode sits at $g^{II} = 0.464$, already far above g^{SS} , reflecting the fact that institutions co-locate researchers who publish across many sources and therefore bridge the gaps that confine the source random walk. Mode 1111 combines both paths and achieves $g^{1111} = 0.347$.

The scatter in the $\mathbf{m} = 1000$ source panel is therefore a consequence of removing the SI – IS mediation that integrates the source network. Without institutional bridging, sources that happen to be the dominant nodes in their local citation neighbourhood are amplified in v relative to the bipartite baseline, while sources whose prestige in the bipartite ranking derives substantially from prestige flowing in via institutional co-authorship links are deflated. The institution panel of Figure 2 is smoother because the institution random walk never relies on a narrow local structure to the same extent: the one-mode projection $\mathbf{M}_I = \mathbf{H}_{IS}\mathbf{H}_{SI}$ connects any two institutions that share sources cited by their authors, which creates a dense, well-connected graph even before the IS – SI paths are included. This asymmetry between sources and institutions is a structural property of the bipartite network and not a feature of the particular corpus or time window.

The bipartite baseline is validated against external journal quality indicators in Figure 3, which plots the four JCR 2024 measures against v in log–log space for the 1,151 sources matched between the baseline ranking and the Web of Science Journal Citation Reports. All four indicators scale approximately as power laws in v : $\text{JIF}_{2024} \propto v^{0.55}$ ($R^2 = 0.52$), $\text{JIF}_{5\text{yr}} \propto v^{0.62}$ ($R^2 = 0.61$), $\text{JCI}_{2024} \propto v^{0.59}$ ($R^2 = 0.58$), and the Article Influence Score $\text{AIS} \propto v^{0.91}$ ($R^2 = 0.82$). AIS, which is itself an eigenvector-based measure constructed from the same citation flows, aligns most closely with v both in slope and goodness of fit. The sub-unity exponents mean that v is more compressed than any of the JCR indicators at the top of the distribution: the bipartite prestige-flow measure redistributes influence more evenly among high-ranking journals than raw citation counts do, consistent with the interpretation that a reference from a prestige-surplus source carries more weight than a reference from a low- v source regardless of volume. The correlation with AIS also confirms that the SI – IS routing through institutions does not distort the basic signal; it refines it by incorporating the institutional dimension of research output.

Figure 4 plots v for all sources and institutions across the five time windows $t_x = 1, \dots, 5$ (2000–2024), with the x -axis locked to the baseline rank. The prestige hierarchy is visibly persistent: the dispersion around the baseline curve narrows as the window approaches the present, and even the most distant window ($t_x = 1$, 2000–04) reproduces the broad shape of the distribution. The Spearman correlation between v in 2000–04 and v in 2020–24 is $\rho_S = 0.663$ for sources and $\rho_S = 0.660$ for institutions on the common sets; for consecutive windows it rises to $\rho_S = 0.905$ (sources) and $\rho_S = 0.864$ (institutions), so the bulk of the apparent instability in Figure 4 is long-run drift rather than short-run volatility. Institutions are no less stable in rank than sources despite having more volatile absolute v values across windows, which reflects the growing size of the institution corpus as additional universities cross the τ_U threshold over the 25-year period. The greater smoothness of the institution panel in Figure 2 and the comparable long-run rank stability in Figure 4 point to the same mechanism: the SI – IS coupling acts as a stabiliser, averaging out the local fluctuations in citation practice that would otherwise produce volatile source rankings.

Figure 5 quantifies the persistence of *prestige-surplus* status—defined as $v > 1$, meaning the unit absorbs more prestige per work than the corpus average—across the full 25-year span. Approximately one quarter of sources and one quarter of institutions are prestige-surplus in each window, and this fraction is stable across time, confirming that $v = 1$ is a structurally consistent threshold rather than a feature of any particular window. Restricting to the 835 sources and 606 institutions present in both the 2000–04 and 2020–24 windows, 83% of sources and 94% of institutions that were prestige-surplus in 2000–04 retain that status in 2020–24. The scatter plots confirm this asymmetry visually: the upper-right quadrant (surplus in both

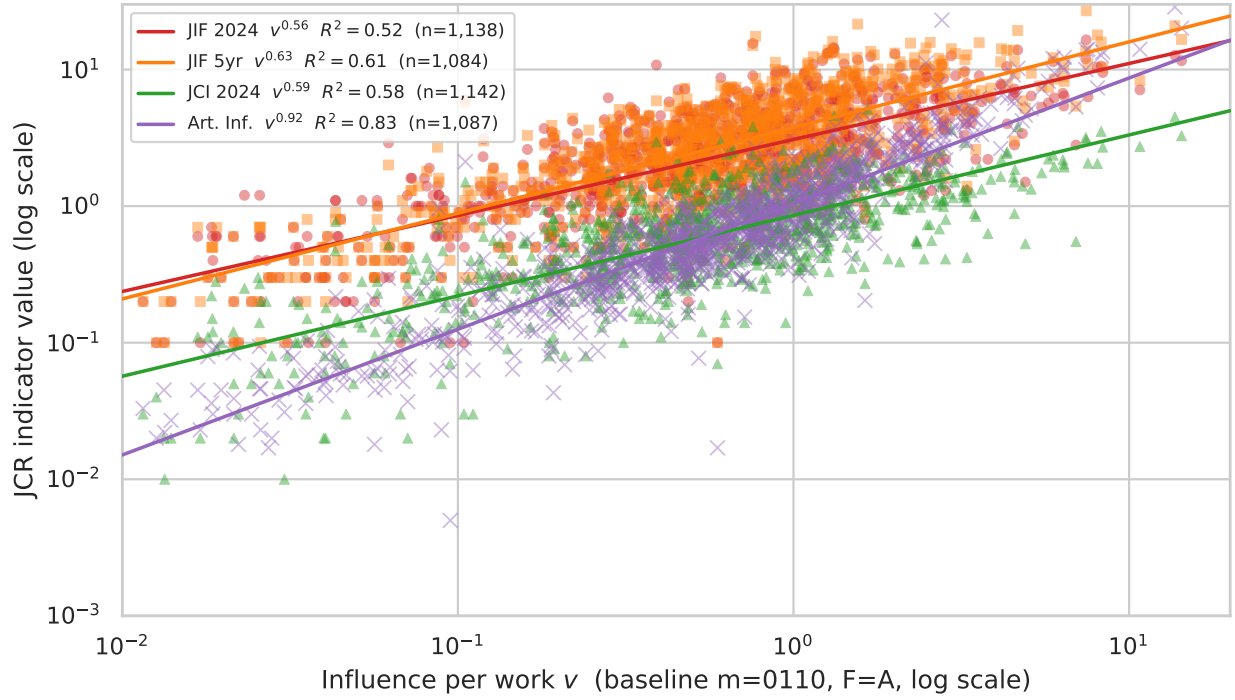


Figure 3: JCR 2024 indicators against prestige per work v (log-log). Scatter: individual sources matched to the Journal Citation Reports ($n = 1,151$). Lines: power-law OLS trend fitted in log-log space; legend reports exponent and R^2 . Dashed line: $v = 1$ (corpus average). Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

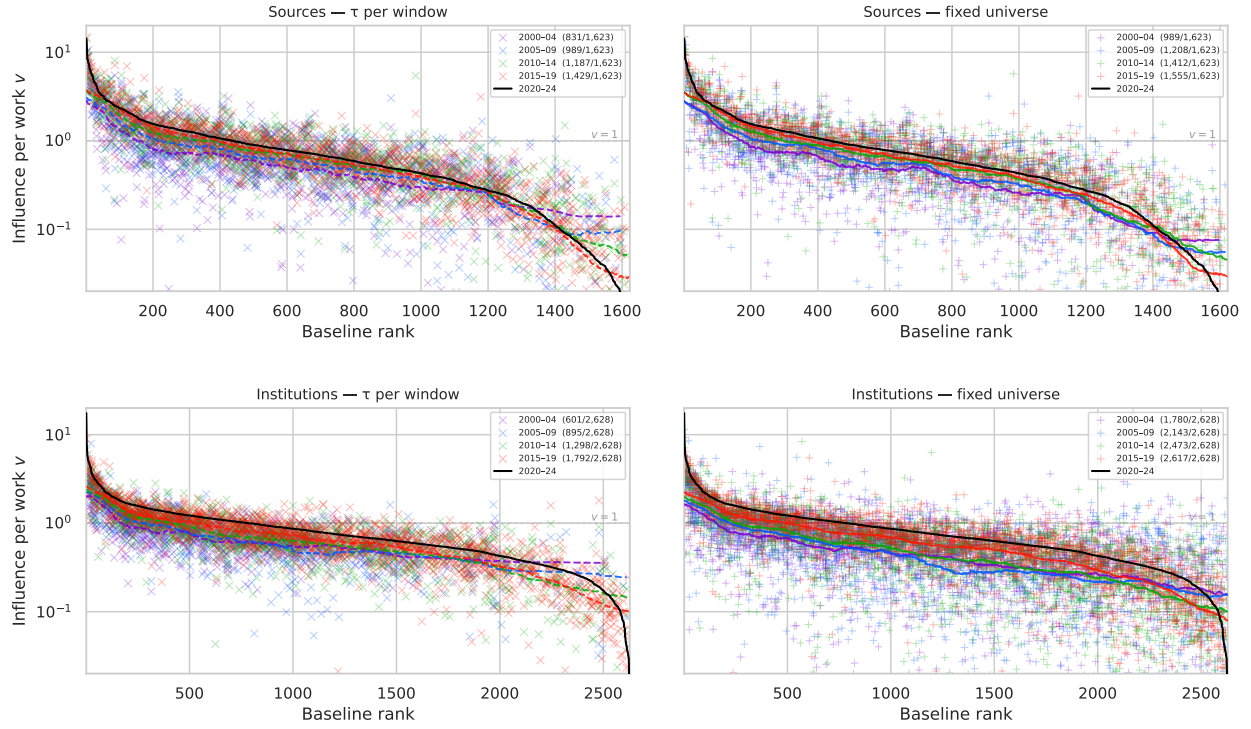


Figure 4: Prestige per work v against baseline rank across five time windows. Top: sources. Bottom: institutions. Black line: baseline $t_x = 5$ (2020–24). Coloured series: $t_x = 1$ (2000–04) through $t_x = 4$ (2015–19). x -axis locked to baseline rank; units absent from a given window are omitted from that series. Parameters: $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

windows) dominates, while the lower-right quadrant (surplus lost) is nearly empty. Classifying each unit's trajectory as a binary sequence (H if $v > 1$, L otherwise) across the five windows shows that monotone paths dominate among institutions (101 monotone rises vs 29 oscillating), while for sources the counts are closer (74 monotone vs 80 oscillating). Once an institution crosses $v = 1$ it almost invariably stays above it; sources are somewhat noisier, but the direction is sustained. The institutions that crossed the threshold between 2000–04 and 2020–24 are predominantly universities in Western countries that expanded their economics and business research output substantially over the period, consistent with the growth of internationally oriented business schools and the associated increase in faculty size.

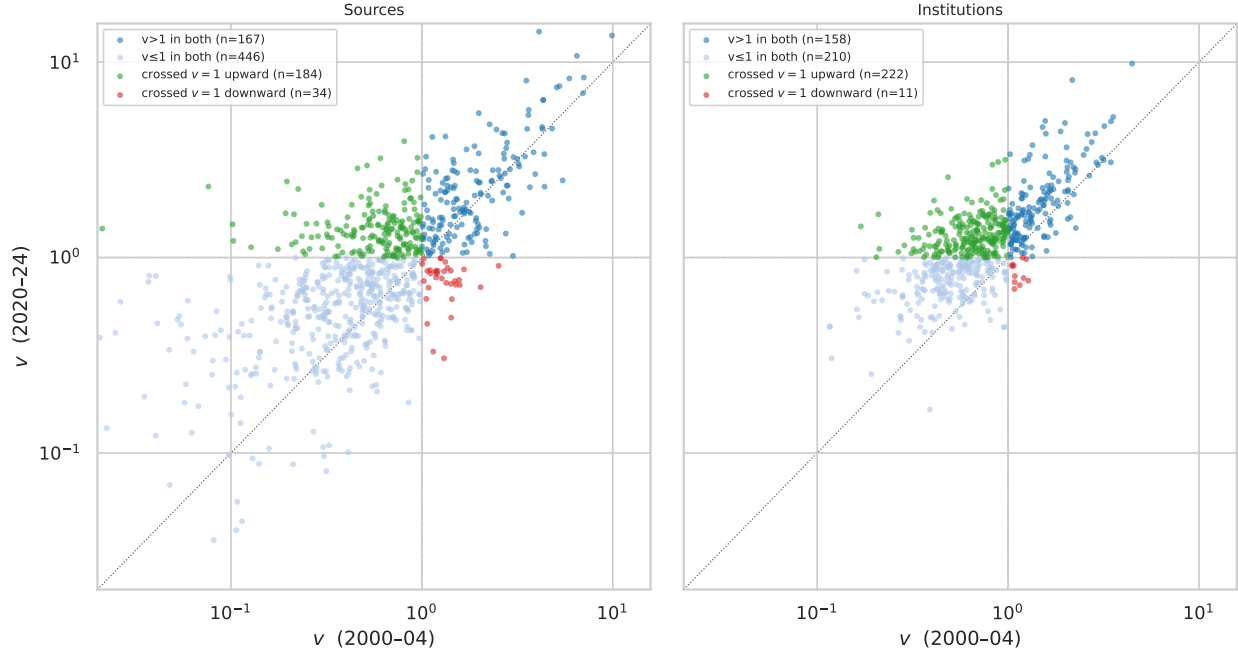


Figure 5: Prestige per work v in 2000–04 against v in 2020–24 for units present in both windows (log–log). Left: sources ($n = 835$). Right: institutions ($n = 606$). Colour indicates threshold-crossing status: blue = prestige-surplus ($v > 1$) in both windows; light blue = below threshold in both; green = crossed $v = 1$ upward; red = crossed $v = 1$ downward. Dotted diagonal: $v_{t_1} = v_{t_5}$. Dashed lines: $v = 1$ on each axis. Parameters: $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

Phase 1: Sources, institutions and fields. Figure 6 shows the prestige-per-work distribution for the baseline and for the field subsets $F = E$ and $F = B$, with the x -axis locked to the baseline rank so that each unit sits at its baseline position and absent units are omitted from that series. Sources at the Economics–Business boundary lose prestige flow from one side under each restriction; the institution panel reflects the same effect through the authorship mapping.

The field shifts in Figure 6 reflect community structure in the citation network: sources cluster into Economics and Business communities that preferentially cite within their own field, and this influences rankings because prestige propagates more readily within a community than across it.

The second eigenpair (λ_2, ϕ_2) of $\mathbf{H}^\top$ characterises this structure precisely. For general α and uniform prior $\mu_p = 1/N$, the solution to (6) has the spectral expansion

$$\boldsymbol{\pi} = (1 - \alpha) \sum_{i \geq 1} \frac{\phi_i^\top \boldsymbol{\mu}}{1 - \alpha \lambda_i} u_i, \quad (13)$$

where (λ_i, u_i, ϕ_i) are the eigenvalues, right eigenvectors, and left eigenvectors of $\mathbf{H}^\top$, ordered $\lambda_1 \geq \lambda_2 \geq \dots$. The $i = 1$ term recovers $\boldsymbol{\pi}^*$ as $\alpha \rightarrow 1$; the $i = 2$ term is the leading community correction, amplified

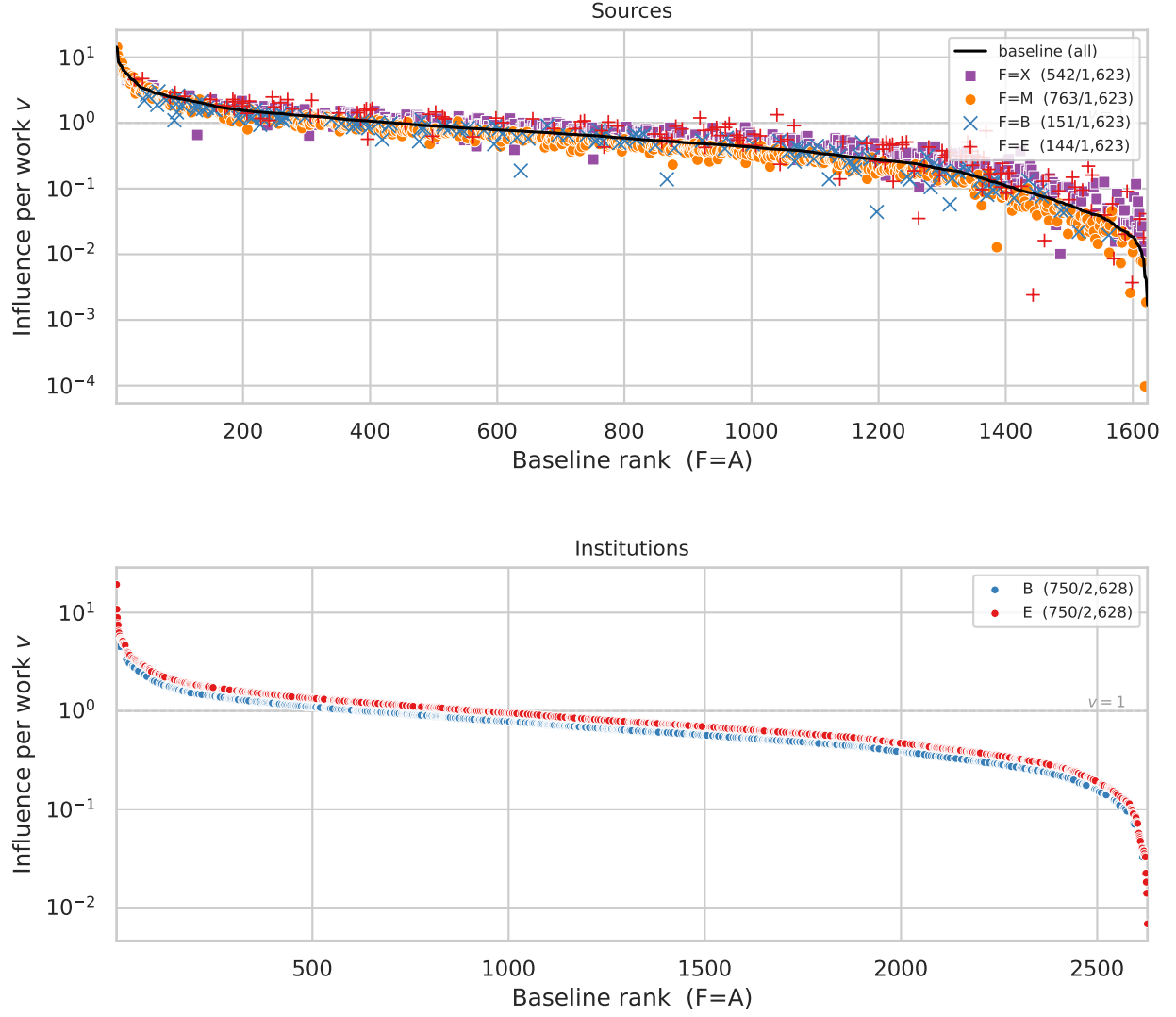


Figure 6: Prestige per work v against baseline rank. Top: sources. Bottom: institutions. Black line: baseline ($F = A$). Red +: $F = E$ ($N_s = [FILL]$). Blue $\times$: $F = B$ ($N_s = [FILL]$). Dashed line: $v = 1$ (corpus average). Parameters: $t_x = 5$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

by $1/(1 - \alpha\lambda_2)$ relative to all higher terms. The spectral gap $g = 1 - \lambda_2$ therefore governs community influence: small g means community membership strongly shapes rankings; large g means rankings are community-neutral. For the SS mode ϕ_2 is expected to separate Economics from Business sources; for II the partition is more diffuse because institutions co-locate faculty from both fields. In the bipartite mode the community operator is the one-mode projection $\mathbf{M}_S = \mathbf{H}_{SI}\mathbf{H}_{IS}$; ϕ_2^S is its second eigenvector and the institution community vector is recovered as $\phi_2^I = \mathbf{H}_{SI}^\top \phi_2^S$ (unit-normalised). Throughout, ϕ_2 is signed so that the unit with the largest work count a_p has $\phi_{2,p} > 0$.

Figure 7 illustrates these eigenvectors by plotting the size-adjusted community measure $\phi_{2,p}/\sqrt{v_p}$ against v_p ; dividing by $\sqrt{v_p}$ removes the normalisation artefact that suppresses $|\phi_{2,p}|$ for low-prestige units.

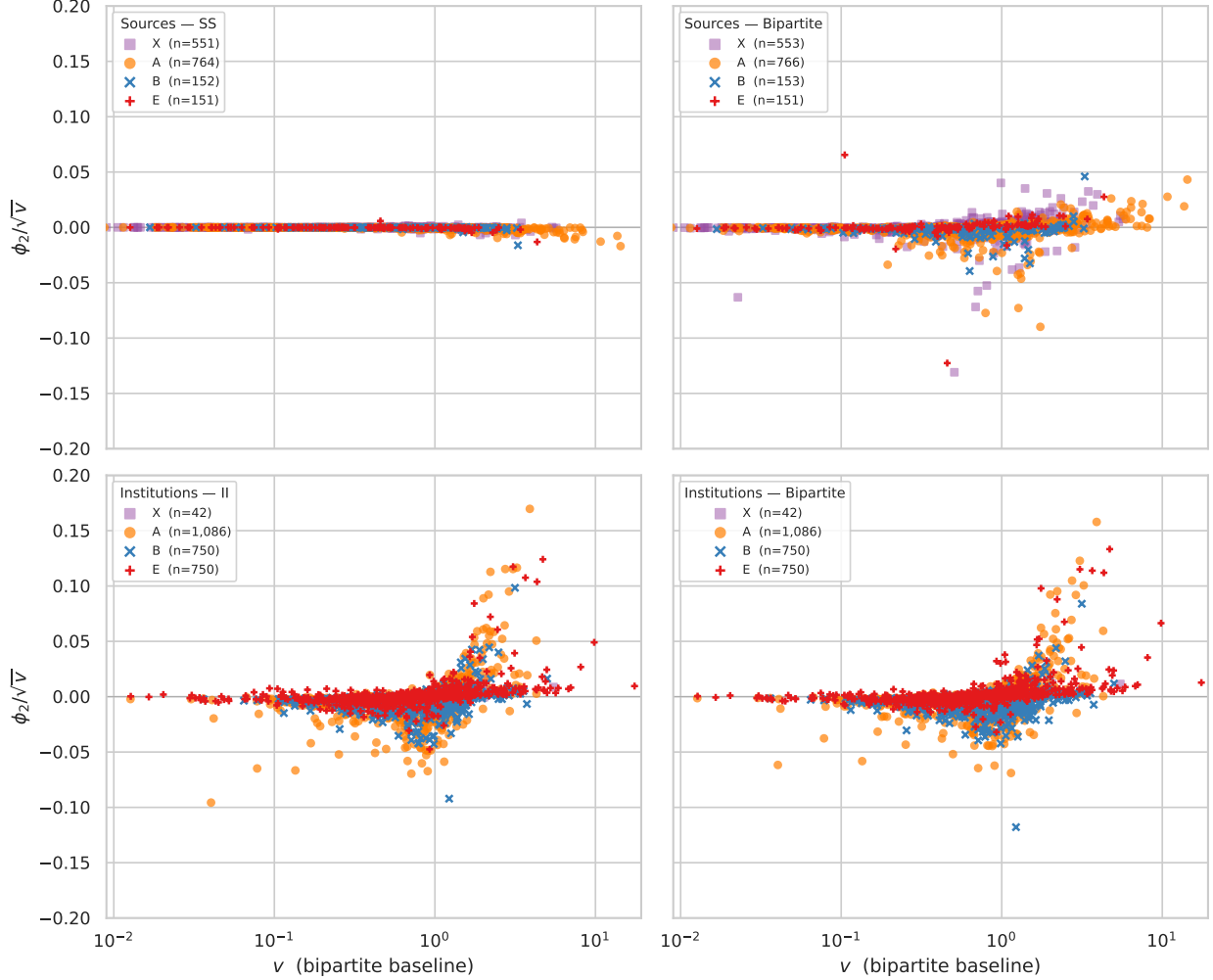


Figure 7: Community measure $\phi_{2,p}/\sqrt{v_p}$ against prestige per work v_p . Top row: sources. Bottom row: institutions. Left column: SS mode $\mathbf{m} = 1000$. Right column: bipartite $\mathbf{m} = 0110$. Sources coloured by field: blue circles = Economics (E); red squares = Business (B); grey crosses = other. Institutions coloured by field-network membership (E-only, B-only, other). x -axis: bipartite baseline v . Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\alpha = 1$.

In the SS source panel (top left), the sign of $\phi_{2,p}/\sqrt{v_p}$ [FILL: cleanly separates Economics (positive) from Business (negative) journals across the full prestige range, with Finance journals concentrated near zero, confirming their role as citation bridges between the two communities]. The bipartite source panel (top right) shows [FILL: a similar partition / greater mixing], with spectral gaps $g^{SS} = [\text{FILL}]$ and $g^{\text{bip}} = [\text{FILL}]$ indicating that routing citation flow through institutions [FILL: partially dissolves / preserves] the community

boundary. The institution panels (bottom row) show a more diffuse partition, consistent with institutions co-locating faculty from both fields; the field-network colouring confirms that E-only institutions [FILL: concentrate on one side / are scattered].

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Addenda

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